



YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training

Workshop - Agriculture AI



Course Objectives

- Gain a comprehensive understanding of the current AI landscape in agriculture, including key technologies and trends.
- Analyze the present situation of AI adoption in agriculture, identifying opportunities and challenges in the World and in Vietnam.
- Explore the provided workshop data, understanding its structure, relevance, and potential applications.
- Clarify the requirements and expectations for the group project, ensuring alignment with workshop goals.
- Develop skills for designing and implementing AI-driven solutions to agricultural challenges.
- Evaluate ethical considerations and responsible AI practices specific to agricultural applications.

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Group Project Guideline

Overview

- This project aims to provide learners with **hands-on experience** in developing an **AI-powered system for agriculture**.
- By utilizing publicly available datasets, learners will **explore the full AI development pipeline**—from data acquisition, preprocessing, and augmentation, to designing, training, and evaluating deep learning models such as Convolutional Neural Networks (CNNs).
- The project will include **performance evaluation** using metrics like accuracy, precision, recall, and F1-score, and culminate in deploying the solution through a user-friendly interface or mobile-ready application.
- The ultimate goal is to create a practical tool that can help farmers identify crop diseases early, reduce yield losses, and support more sustainable and efficient agricultural practices.

Datasets

Learners may either **use a dataset of their choice** or select from one of the recommended datasets listed below:

PlantVillage dataset

Dataset of diseased plant leaf images and corresponding labels

Multi-class Weather

This dataset contains images of weather suitable for weather analysis

Diseases on Citrus

The dataset contains an image gallery of healthy and unhealthy citrus fruits and leaves

FAOSTAT

This dataset of various statistics from Food and Agriculture Organization of the UN (FAO-UN)

GBIF dataset

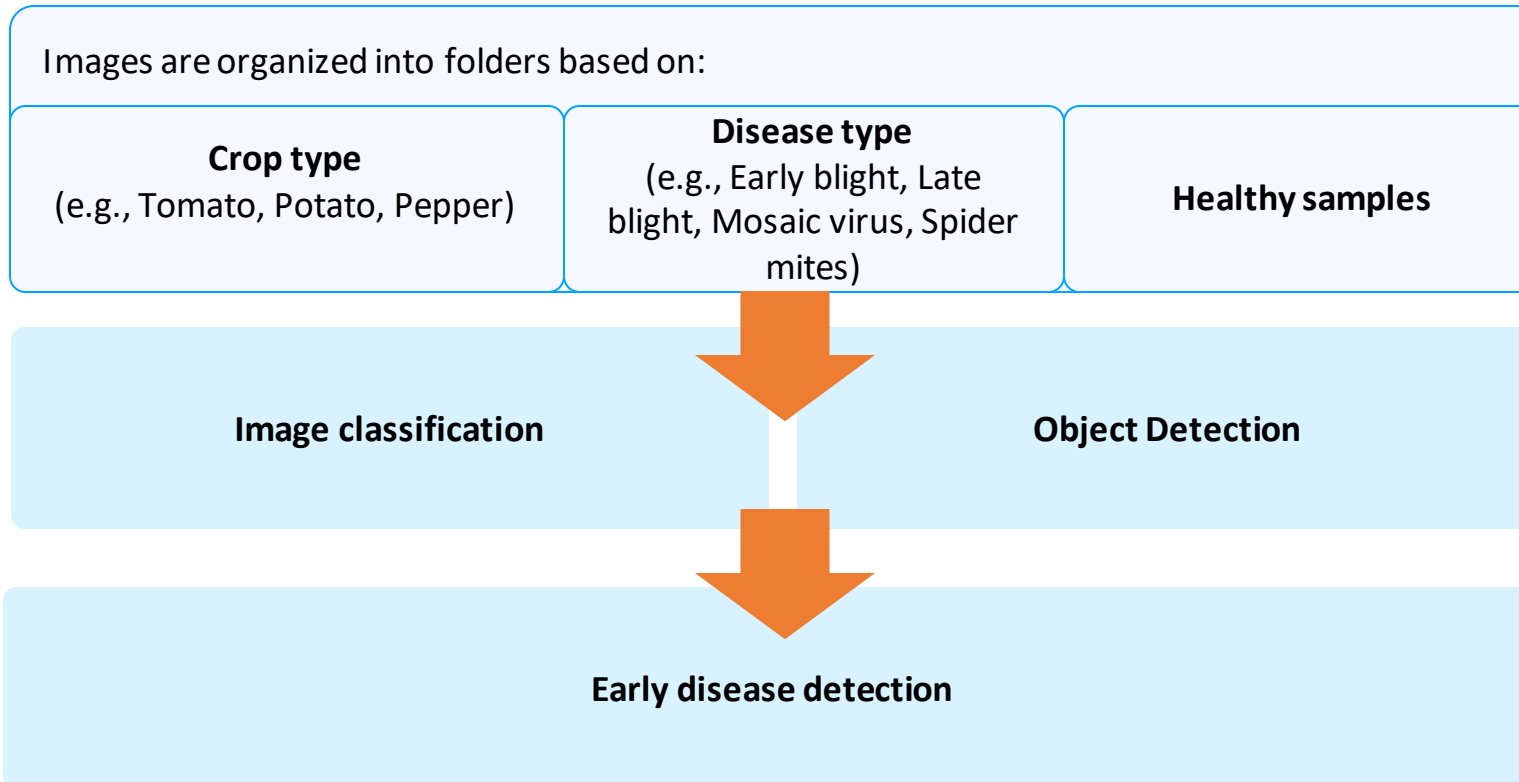
International network and data infrastructure funded by the world's governments.

Dataset

PlantVillage dataset

<https://www.kaggle.com/datasets/emmarex/plantdisease>

Image-based dataset featuring diseased and healthy plant leaves



Tomato Late blight



Tomato Spider mite



Pepper Bacterial spot



Pepper Healthy

Dataset

Multi-class Weather

<https://data.mendeley.com/datasets/4drtyfjtfy/1>

Multi-class weather dataset(MWD) for image classification is a valuable dataset used in the research paper entitled “Multi-class weather recognition from still image using heterogeneous ensemble method”. The dataset provides a platform for outdoor weather analysis by extracting various features for recognizing different weather conditions.

The disease targeted in the data sets are:

- **cloudy*.jpg**: Cloudy sky
- **rain*.jpg** : rainy sky
- **shine*.jpg** : sunny sky
- **sunrise*.jpg**: sun rise sky

Image classification

Weather detection for irrigation management or weather tracking



Cloudy



Rain



Shine



Sun rise

Dataset

Disease on Citrus Fruit and Leaves

<https://data.mendeley.com/datasets/3f83gxm57/1>

The dataset contains an image gallery of healthy and unhealthy citrus fruits and leaves that could be usable for the researchers to prevent plants from diseases using advanced computer vision techniques. The disease targeted in the data sets are the Blackspot, Canker, Scab, Greening, and Melanose.



Greening



Canker



Black Spot



Scab

Images are organized by name:

Part type
(e.g., Fruit and Leaves)

Disease type
(e.g., Healthy, Black spot, Canker, Greening and Scab)

Image classification

Object Detection

Early disease detection

Dataset

FAOSTAT (multiple datasets)

<https://www.fao.org/faostat/en/#home>

FAOSTAT provides free access to food and agriculture data for over 245 countries and territories and covers all FAO regional groupings from 1961 to the most recent year available.

Various statistics as table (time series/non-time series)

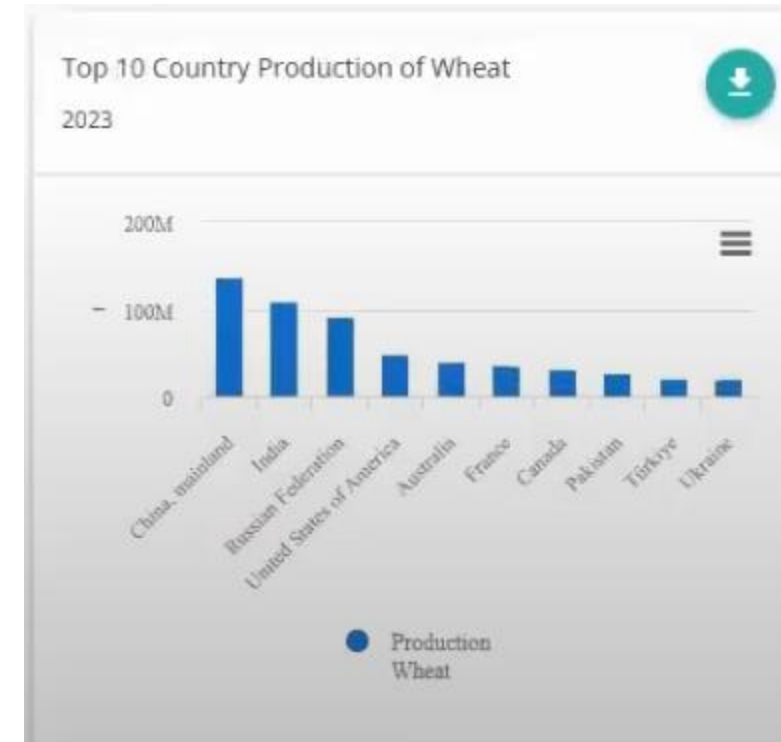
Descriptive

Diagnostic

Predictive

Prescriptive

Analyse and tell stories about agriculture statistics



Wheat production by country data

Dataset

GBIF | Global Biodiversity Information Facility (multiple datasets)

<https://www.gbif.org/>

GBIF—the Global Biodiversity Information Facility—is an international network and data infrastructure funded by the world's governments and aimed at providing anyone, anywhere, open access to data about all types of life on Earth.

You can find various dataset here.

Various statistics as table (time series/non-time series) or list of plants/animals

Descriptive

Diagnostic

Predictive

Prescriptive

Analyse and tell stories about agriculture statistics



Global soil organisms

Project Guideline

Your final report should be structured according to the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** standard:

1

Business Understanding

Define the problem

Understand pain points & Identify goals

2

Data Understanding

Explore and explain the dataset

Note external factors

3

Data Preparation

Clean & preprocess the data

Apply data augmentation

4

Modeling

Choose model type and train model

Tune hyperparameters

5

Evaluation

Use standard metrics to compare models

Analyze misclassifications and confusion

6

Deployment/Delivery

Visualize detection results and insights

Suggest future improvements

Conclusion

Conclusion

- **Critical Role of AI:** AI is transforming agriculture by enhancing productivity, optimizing resource use, and addressing challenges like climate change and food security, support for sustainable farming.
- **Immense Potential:** AI-driven solutions, such as precision farming, disease prediction, and smart irrigation, can revolutionize crop yields and reduce environmental impact, particularly in Vietnam's agriculture sector.
- **Understand the CRISP-DM (Cross-Industry Standard Process for Data Mining)**
- **Skill Development Needs:** Expanding expertise in AI model (e.g., CNN, RNN) and data analysis is essential for farmers effectively implement AgriTech solutions.:
 - Knowing the full process to develop an AgriTech solution
 - Proficiency in AI techniques (Machine Learning, Deep Learning, NLP, Generative AI) and skills like image embedding, augmentation, and fine-tuning models for specific tasks, and connecting modalities.
 - Business understanding of agriculture and how we can do exploit data for AI solutions.